

Unit 6: Multivariable Calculus Foundations

Prepared by Staples Education

Condensed Cheat Sheet

Unit 6 at a Glance: Core Sub-Topics

Differential equations in more than one variable need the language of multivariable calculus. This unit builds that bridge: we learn to differentiate one variable at a time, to package those partials into a gradient, to chain through intermediate variables, to locate and classify extrema, and to read the geometry of level curves — the same geometry that decides when a differential $M dx + N dy$ is *exact*.

- **Partial Derivatives** — differentiate in one variable while holding the others constant; f_x, f_y .
- **Higher-Order and Mixed Partial**s — Clairaut's theorem: $f_{xy} = f_{yx}$ when the second partials are continuous.
- **Gradient and Directional Derivatives** — ∇f collects the partials; $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ measures change along a unit direction.
- **The Multivariable Chain Rule** — sum one product term per path from the output down to the independent variable.
- **Critical Points and Optimization** — set $\nabla f = \mathbf{0}$, then classify with the second-derivative (Hessian) test.
- **Level Curves and Exactness** — $f(x, y) = c$, ∇f normal to the curve, and the exact-differential test $\partial_y M = \partial_x N$.

Part I — Condensed Cheat Sheet

6.1 Partial Derivatives

A partial derivative differentiates a multivariable function in *one* variable while treating every other variable as a constant symbol. It measures the instantaneous rate of change of f along a single coordinate axis.

Definition of the Partial Derivatives

$$f_x = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}, \quad f_y = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

- To compute f_x , **hold y constant** and differentiate in x with the ordinary rules; any pure- y term is a constant and dies.
- Holding a variable *constant* is not the same as *setting it to zero*: the symbol stays in the answer.

Worked Example 6.1: First partials of $f = x^2y + y^3$

Treat y as constant for f_x , then x as constant for f_y :

$$f_x = 2xy + 0 = 2xy, \quad f_y = x^2 + 3y^2.$$

The y^3 term vanishes under ∂_x , while the x^2 term contributes its coefficient x^2 under ∂_y .

6.2 Higher-Order and Mixed Partial

Each first partial can itself be differentiated again, giving four second-order partials: two *pure* (f_{xx}, f_{yy}) and two *mixed* (f_{xy}, f_{yx}). Clairaut's theorem says the mixed pair agrees whenever the function is smooth enough.

Clairaut's Theorem on Mixed Partial

If the second partial derivatives of f are **continuous** near a point, then the mixed partials are equal there:

$$f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = f_{yx}.$$

The order of differentiation does not matter — a function of two variables has **four** second partials but only **three** distinct values.

Worked Example 6.2: Mixed partials of $f = x^2y^3$

Differentiate in x first, then in y — and the reverse:

$$f_x = 2xy^3 \Rightarrow f_{xy} = 6xy^2, \quad f_y = 3x^2y^2 \Rightarrow f_{yx} = 6xy^2.$$

Both routes land on $6xy^2$, exactly as Clairaut guarantees.

6.3 The Gradient and Directional Derivatives

The gradient packages the partials into a single vector that points in the direction of steepest increase. Dotting it with a *unit* direction gives the rate of change along that direction — the directional derivative.

Gradient, Directional Derivative, and Steepest Ascent

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right), \quad D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} \quad (\mathbf{u} \text{ a unit vector}).$$

- The **maximum** rate of increase at a point is $|\nabla f|$, attained in the direction $\mathbf{u} = \nabla f / |\nabla f|$.
- ∇f is **perpendicular** to the level curve through the point.

Common Pitfall: The Direction Must Be a Unit Vector

$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ is only valid when $\mathbf{u}$ is **normalized**. Given a raw direction $\mathbf{v}$, divide first: $\mathbf{u} = \mathbf{v}/|\mathbf{v}|$. Skipping the normalization rescales the answer by $|\mathbf{v}|$ and silently corrupts every directional derivative and max-rate computation that follows.

Worked Example 6.3: Directional derivative of $f = x^2 + xy + y^2$

At $(1, 2)$, $\nabla f = (2x + y, x + 2y) = (4, 5)$. Along $\mathbf{v} = (3, 4)$, normalize to $\mathbf{u} = (\frac{3}{5}, \frac{4}{5})$:

$$D_{\mathbf{u}}f = (4, 5) \cdot \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{12}{5} + \frac{20}{5} = \frac{32}{5} = 6.4.$$

The steepest ascent there points along $(4, 5)$ with rate $|\nabla f| = \sqrt{41} \approx 6.40$.

6.4 The Multivariable Chain Rule

When the inputs of f are themselves functions, differentiate by summing one product term for every path from the output down to the variable you are differentiating with respect to. A small *tree diagram* bookkeeps the paths.

Chain Rule Trees: Counting the Paths

One independent variable $z = f(x(t), y(t))$ — two paths to t :

$$\frac{dz}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \nabla f \cdot (x', y').$$

Two independent variables $z = f(x(s, t), y(s, t))$ — two paths to each:

$$\frac{\partial z}{\partial s} = f_x x_s + f_y y_s, \quad \frac{\partial z}{\partial t} = f_x x_t + f_y y_t.$$

1. Draw a branch from z to each intermediate variable (x, y) .
2. Extend a branch from each intermediate variable to the target (t) .
3. **Multiply** along each branch, then **add** the branches.

Worked Example 6.4: Chain rule for $z = x^2y$, $x = t^2$, $y = t^3$

With $z_x = 2xy$, $z_y = x^2$, $x' = 2t$, $y' = 3t^2$:

$$\frac{dz}{dt} = (2xy)(2t) + (x^2)(3t^2) = (2t^5)(2t) + (t^4)(3t^2) = 4t^6 + 3t^6 = 7t^6.$$

The direct substitution $z = t^4 \cdot t^3 = t^7$ confirms $dz/dt = 7t^6$.

6.5 Critical Points and Optimization

Interior extrema of a smooth $f(x, y)$ occur where the gradient vanishes. The *second-derivative test* then uses the sign of the Hessian discriminant to sort each critical point into a minimum, maximum, or saddle.

Locating and Classifying Critical Points

1. **Solve** $\nabla f = \mathbf{0}$, i.e. $f_x = 0$ and $f_y = 0$ simultaneously, to find all critical points.
2. **Form** the discriminant $D = f_{xx}f_{yy} - (f_{xy})^2$ at each point.
3. **Classify** with D and f_{xx} (Step in the box below).

The Second-Derivative (Hessian) Test

$$D = f_{xx} f_{yy} - (f_{xy})^2.$$

- $D > 0$, $f_{xx} > 0$: **local minimum**.
- $D > 0$, $f_{xx} < 0$: **local maximum**.
- $D < 0$: **saddle point** (no extremum).
- $D = 0$: the test is **inconclusive**.

Common Pitfall: Reading the Sign Boundary Wrong

The sign of f_{xx} only decides min vs. max *after* $D > 0$ confirms an extremum. When $D < 0$ the point is a **saddle** regardless of f_{xx} , and when $D = 0$ the test says *nothing* — do not declare a saddle by default. A single mishandled sign flips a minimum into a maximum.

6.6 Level Curves, Gradients, and Exactness

Level curves $f(x, y) = c$ slice a surface into contour lines; the gradient cuts across them at right angles. This same gradient structure is exactly what makes a first-order differential *exact* — the payoff that carries into Unit 7.

Level Curves and the Exact-Differential Test

A **level curve** is the set $\{(x, y) : f(x, y) = c\}$; ∇f is normal to it. A differential $M(x, y) dx + N(x, y) dy$ is **exact** on a simply connected region exactly when

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

in which case it equals df for a potential f with $f_x = M$, $f_y = N$ — and the field $\mathbf{F} = (M, N) = \nabla f$ is **conservative**.

Common Pitfall: Path Dependence and the Wrong Cross Partial

The exactness/conservative test compares the *cross* partials $\partial_y M$ and $\partial_x N$ — not $\partial_x M$ and $\partial_y N$. When they differ the field is path-dependent: $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends on the route, so there is *no* single potential to read off.

Unit 6 Key Takeaway

Multivariable calculus is the toolkit the rest of the course runs on. The **partials** f_x, f_y collect into the **gradient** ∇f , which both gives directional derivatives $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ and points along steepest ascent. The **chain rule** sums one term per path; **optimization** sets $\nabla f = \mathbf{0}$ and classifies with $D = f_{xx}f_{yy} - f_{xy}^2$; and the exactness test $\partial_y M = \partial_x N$ is the gradient idea in disguise, the gateway to exact equations.